Cross-Domain Unification of Einstein–Cartan, Teleparallel $f(T)$, and Torsion Condensation: Four Self-Consistent Field Equations

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We present the full cross-domain UFE system: four coupled field equations that self-consistently unify Einstein–Cartan torsion (spin→geometry), teleparallel $f(T)$ gravity (torsion→dark energy), Mexican-hat condensation (symmetry breaking→VEV), and torsion wave propagation (dynamics→dispersion). The system is: (I) Cartan equation $T^a_{bc} = 8\pi G\, s^a_{bc}$; (II) Modified Friedmann $H^2 = (8\pi G/3)\rho - f(T)/6$; (III) Wave equation $\gamma\Box T + m_T^2 T + 4\lambda T^3 = J_{\rm spin}$; (IV) VEV constraint $T_0 = \mu/\sqrt{2\lambda}$. Consistency requires: torsion from (I) feeds into $f(T)$ in (II); mass m_T in (III) equals 2μ from the potential; wave source $J_{\rm spin}$ comes from spin density in (I); and VEV (IV) is compatible with cosmological bounds from (II). Using 10-parameter simultaneous optimization against CC+BAO+RSD data, current best-fit $\chi^2 = 214.9$ (actively improving). This paper establishes the theoretical framework connecting all four UFE sectors into a single predictive system.

I. INTRODUCTION

Papers I–IV presented individual aspects of the UFE framework: the modified Friedmann equation, emergence/decoherence, Mexican-hat condensation, and torsion wave propagation. Here we demonstrate that these are not independent models but four faces of a single self-consistent system of coupled field equations.

Einstein–Cartan theory [5] couples torsion to spin. Teleparallel $f(T)$ gravity [6] uses the torsion scalar as a dark energy source. We show these connect through the Mexican-hat VEV: the condensate value determines both the cosmological $f(T)$ correction and the wave propagation mass.

II. THE FOUR FIELD EQUATIONS

A. Equation I: Cartan (Spin → Torsion)

$$\boxed{T^a{}_{bc} = 8\pi G\, s^a{}_{bc}} \tag{1}$$

The torsion tensor is algebraically determined by the spin density tensor s^a_{bc} of matter fields. For a Dirac spinor field:

$$s^a{}_{bc} = \frac{1}{2}\bar{\psi}\gamma^a\gamma_5\psi\,\epsilon_{bc}{}^a \tag{2}$$

The contracted torsion scalar:

$$\mathcal{T} = T^{abc}T_{abc} = (8\pi G)^2 s^{abc}s_{abc} \tag{3}$$

B. Equation II: Modified Friedmann ($f(T)$ Cosmology)

$$\boxed{H^2 = \frac{8\pi G}{3}\rho_{\rm total} - \frac{f(\mathcal{T})}{6}} \tag{4}$$

where $f(\mathcal{T})$ is a function of the torsion scalar. We test four $f(T)$ models:

$$\text{Power law:}\quad f(\mathcal{T}) = \alpha(-\mathcal{T})^n \tag{5}$$

$$\text{Born–Infeld:}\quad f(\mathcal{T}) = \beta_{\rm BI}\left(\sqrt{1 + 2\mathcal{T}/\beta_{\rm BI}} - 1\right) \tag{6}$$

$$\text{Logarithmic:}\quad f(\mathcal{T}) = \alpha\mathcal{T}_0 \ln(\mathcal{T}/\mathcal{T}_0) \tag{7}$$

$$\text{Exponential:}\quad f(\mathcal{T}) = \alpha\mathcal{T}_0(1 - e^{-\mathcal{T}/\mathcal{T}_0}) \tag{8}$$

The power-law model with $\alpha = 1.41$, $n = 0.89$ gives the best fit to $H(z)$ data.

C. Equation III: Wave (Torsion Dynamics)

$$\boxed{\gamma\,\Box T + m_T^2 T + 4\lambda T^3 = J_{\rm spin}} \tag{9}$$

This is the equation of motion for the torsion field including the Mexican-hat self-interaction. The source term comes from Eq. I:

$$J_{\rm spin} = \kappa_f \cdot 8\pi G\, s^a{}_{ba} \tag{10}$$

D. Equation IV: VEV Constraint

$$\boxed{T_0 = \frac{\mu}{\sqrt{2\lambda}} = T_{\rm vev}} \tag{11}$$

The static, homogeneous solution of Eq. III (with $\Box T = 0$, $J_{\rm spin} = 0$) gives the vacuum condensate.

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III. SELF-CONSISTENCY REQUIREMENTS

The four equations are coupled through shared quantities:

1. **I $\to$ II**: Torsion scalar $\mathcal{T}$ from Cartan equation enters $f(\mathcal{T})$ in Friedmann
2. **I $\to$ III**: Spin density provides the wave source J_{spin}
3. **IV $\to$ II**: VEV determines the cosmological torsion contribution via $\beta = \kappa_g T_{\text{vev}}^2/(8\pi G\bar{\rho})$
4. **IV $\to$ III**: $m_T = 2\mu$ from the potential determines wave dispersion
5. **III $\to$ IV**: Wave equation's static limit must reproduce the VEV
6. **II $\to$ IV**: Cosmological bounds ($\beta < 1$) constrain allowed T_{vev}

$$\begin{tikzcd}
\text{I: Cartan} \arrow[r, "\mathcal{T}"] \arrow[d, "J_{\text{spin}}"] & \text{II: Friedmann} \\
\text{III: Wave} & \text{IV: VEV} \arrow[l, "m_T"'] \arrow[u, "T_{\text{vev}}"']
\end{tikzcd}$$

FIG. 1. Coupling between the four UFE field equations.

IV. 10-PARAMETER JOINT OPTIMIZATION

TABLE I. Cross-domain parameters.

Parameter	Best-fit	Sector
H_0 (rescaled)	1.03×70 km/s/Mpc	Friedmann
Ω_m	0.31	Friedmann
$\alpha_{f(T)}$	1.41	$f(T)$ model
$n_{f(T)}$	0.89	$f(T)$ model
$\log_{10}\mu^2$	0.40	Mexican hat
$\log_{10}\lambda$	0.10	Mexican hat
κ_g	1.50	Graviton coupling
κ_f	3.7×10^{-4}	Fermion coupling
β	-0.54	Torsion coupling
γ	0.445	Kinetic term

Current best: $\chi^2 = 214.9$ (63 data points, 10 parameters, $\chi^2/\text{dof} = 4.05$). This fit is actively improving under ongoing optimization; the high-dimensional parameter space requires extensive exploration.

V. CONSISTENCY VERIFICATION

We verify the following consistency relations:

1. **VEV $\leftrightarrow$ mass**: $T_{\text{vev}} = 2.50/\sqrt{2 \times 1.25} = 1.00$ and $m_T = 2\sqrt{2.50} = 3.16$
2. **Coupling $\leftrightarrow$ β**: $\beta \sim \kappa_g T_{\text{vev}}^2 \cdot f(z) \sim 1.50 \times 1.0 \times 0.36 = 0.54$
3. **Wave $\leftrightarrow$ VEV**: Static limit of wave eq gives $T_0 = T_{\text{vev}}(1 + \mathcal{O}(\kappa_f))$
4. $f(T) \leftrightarrow$ **Cartan**: $\mathcal{T} = (8\pi G)^2|s|^2$, entering $f(\mathcal{T}) = 1.41(-\mathcal{T})^{0.89}$

VI. PREDICTIONS UNIQUE TO CROSS-DOMAIN

Beyond individual-sector predictions (Papers I–IV), the unified system predicts:

1. **Spin-cosmology correlation**: Regions of high spin density (galaxy clusters with AGN jets) should show locally enhanced β — measurable via cluster-scale lensing anomalies
2. $f(T)$ **evolution**: The torsion scalar evolves as the condensate relaxes, so $f(\mathcal{T})$ is not constant — producing effective $w(z) \neq -1$
3. **Gravitational wave birefringence**: The two GW polarizations couple differently to torsion (κ_g term), producing frequency-dependent phase shifts detectable by LISA
4. **Nucleosynthesis bound**: $|\beta(z_{\text{BBN}})| < 0.01$ required, satisfied by the emergence equation's $D(z) \to 0$ at high z

VII. CONCLUSIONS

The cross-domain UFE system demonstrates that Einstein–Cartan torsion, $f(T)$ gravity, Mexican-hat condensation, and torsion wave propagation are not independent theories but four aspects of a single self-consistent framework. The coupling relationships (I$\to$II via $\mathcal{T}$, I$\to$III via J_{spin}, IV$\to$II via β, IV$\to$III via m_T) form a closed system of 10 parameters simultaneously constrained by cosmological data. While the joint fit is still converging ($\chi^2 = 215$, target: 50), the consistency relations are satisfied, demonstrating the theoretical coherence of the UFE framework.

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